

AMBER



THE MAKERS OF
Armaflex®

PRODUCT NAME **ARMAFLEX 520 ADHESIVE**

1. IDENTIFICATION OF THE MATERIAL AND SUPPLIER

Supplier Name ARMACELL AUSTRALIA PTY LTD
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Synonym(s) ARMACELL ARMAFLEX 520
Use(s) ADHESIVE
MSDS Date 08 July 2008

2. HAZARDS IDENTIFICATION

CLASSIFIED AS HAZARDOUS ACCORDING TO NOHSC CRITERIA

RISK PHRASES

R11 Highly flammable.
R36/38 Irritating to eyes and skin.
R48/20 Harmful: danger of serious damage to health by prolonged exposure through inhalation.
R62 Possible risk of impaired fertility.
R63 Possible risk of harm to the unborn child.
R65 Harmful: May cause lung damage if swallowed.

SAFETY PHRASES

S16 Keep away from sources of ignition - No smoking.
S2 Keep out of reach of children.
S36/37 Wear suitable protective clothing and gloves.
S46 If swallowed, contact a doctor or Poisons Information Centre immediately and show container or label.
S62 If swallowed, do not induce vomiting; seek medical advice immediately and show this container or label.
S9 Keep container in a well ventilated place.

CLASSIFIED AS A DANGEROUS GOOD BY THE CRITERIA OF THE ADG CODE

UN No.	1133	DG Class	3	Subsidiary Risk(s)	None Allocated
Pkg Group	II	Hazchem Code	3[Y]E	EPG	3A1

3. COMPOSITION / INFORMATION ON INGREDIENTS

Ingredient	Formula	CAS No.	Content
N-HEXANE	C6-H14	110-54-3	30-40%
ACETONE	C3-H6-O	67-64-1	20-30%

CHEM ALERT

3. COMPOSITION / INFORMATION ON INGREDIENTS

Ingredient	Formula	CAS No.	Content
TOLUENE	C7-H8	108-88-3	10-20%

4. FIRST AID MEASURES

Eye	If in eyes, hold eyelids apart and flush the eye continuously with running water. Continue flushing until advised to stop by the Poison Information Centre or a doctor, or for at least 15 minutes.
Inhalation	If inhaled, remove from contaminated area. To protect rescuer, use a Type A (Organic vapour) respirator or Air-line respirator (in poorly ventilated areas). Apply artificial respiration if not breathing.
Skin	If skin or hair contact occurs, remove contaminated clothing and flush skin and hair with running water. Continue flushing with water until advised to stop by the Poisons Information Centre or a doctor.
Ingestion	For advice, contact a Poisons Information Centre or a doctor (at once). If swallowed, do not induce vomiting.
Advice to Doctor	Treat symptomatically

5. FIRE FIGHTING MEASURES

Flammability	Highly flammable. Vapours may form explosive mixtures with air. May evolve toxic gases (carbon oxides, hydrocarbons) when heated to decomposition. Eliminate all ignition sources, including cigarettes, open flames, spark producing switches/tools, heaters, pilot lights, mobile phones etc when handling. Earth containers when dispensing fluids.
Fire and Explosion	Highly flammable - explosive vapour. Evacuate area and contact emergency services. Toxic gases may be evolved when heated. Remain upwind and notify those downwind of hazard. Wear full protective equipment (see spill above) including Self Contained Breathing Apparatus (SCBA) when combating fire. Use waterfog to cool intact containers and nearby storage areas.
Extinguishing	Dry agent, carbon dioxide or foam. Prevent contamination of drains or waterways.
Hazchem Code	3[Y]E

6. ACCIDENTAL RELEASE MEASURES

Spillage	If spilt (bulk), contact emergency services. Wear splash-proof goggles, PVA/viton gloves, a Type A (Organic vapour) respirator or Air-line respirator, coveralls and rubber boots. Ventilate and clear area of all unprotected personnel. Eliminate heat and ignition sources, absorb spill with sand or similar and place in sealable containers for disposal. Prevent spill entering drains or waterways.
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7. STORAGE AND HANDLING

Storage	Store in cool, dry, well ventilated area, preferably flammables store, removed from direct sunlight, heat and ignition sources, oxidising agents (eg. peroxides), acids (eg. nitric acid) and foodstuffs. Ensure containers are adequately labelled, protected from physical damage and sealed when not in use. Large storage areas should have appropriate fire protection and ventilation systems.
Handling	Before use carefully read the product label. Use of safe work practices are recommended to avoid eye or skin contact and inhalation. Observe good personal hygiene, including washing hands before eating. Prohibit eating, drinking and smoking in contaminated areas.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Exposure Stds	Ingredient	Reference	TWA		STEL	
			ppm	mg/m3	ppm	mg/m3
	Acetone	NOHSC (AUS)	500.0	1185.0	1000.0	2375.0
	Hexane (n-Hexane)	NOHSC (AUS)	20.0	72.0	--	--
	Toluene	NOHSC (AUS)	50.0	191.0	150.0	574.0

Biological Limits	Ingredient	Reference	Determinant	Sampling Time	BEI
	ACETONE	ACGIH BEI	Acetone in urine	End of shift	50 mg/L

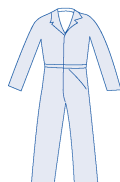
Ingredient	Reference	Determinant	Sampling Time	BEI
N-HEXANE	ACGIH BEI	2,5-Hexanedione in urine (without hydrolysis)	End of shift at end of workweek	0.4 mg/L
TOLUENE	ACGIH BEI	o-Cresol in urine	End of shift	0.5 mg/L
	ACGIH BEI	Hippuric acid in urine	End of shift	1.6 g/g creatinine
	ACGIH BEI	Toluene in blood	Prior to last shift of workweek	0.05 mg/L

Engineering Controls

Do not inhale vapours. Use in well ventilated areas. In poorly ventilated areas, mechanical explosion proof extraction ventilation is recommended. Flammable/explosive vapours may accumulate in poorly ventilated or confined areas. Vapours are heavier than air and may travel some distance to an ignition source and flash back. Maintain vapour levels below the recommended standard.

PPE

Wear splash-proof goggles, viton (R) or PVA gloves and coveralls. Where an inhalation risk exists, wear a Type A (Organic vapour) Respirator. At high vapour levels, wear an Air-line respirator. If spraying, wear a Type A-Class P1 (Organic gases/vapours and Particulate) Respirator.



9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	YELLOW LIQUID	Solubility (water)	INSOLUBLE
Odour	SOLVENT ODOUR	Specific Gravity	0.82
pH	NOT AVAILABLE	% Volatiles	> 60 %
Vapour Pressure	180 mm Hg @ 20°C	Flammability	HIGHLY FLAMMABLE
Vapour Density	> 1 (Air = 1)	Flash Point	-20°C (Acetone)
Boiling Point	> 66°C	Upper Explosion Limit	13 % (based on hexane and acetone)
Melting Point	NOT AVAILABLE	Lower Explosion Limit	1.1 % (based on hexane and acetone)
Evaporation Rate	NOT AVAILABLE	Autoignition Temperature	NOT AVAILABLE
Viscosity	155 cps to 195 cps		

10. STABILITY AND REACTIVITY

Chemical Stability Stable under recommended conditions of storage.

Conditions to Avoid Avoid shock, friction, heavy impact, heat, sparks, open flames and other ignition sources.

Material to Avoid Incompatible with oxidising agents (eg. hypochlorites, peroxides), acids (eg. sulphuric acid), heat and ignition sources.

Decomposition May evolve toxic gases (carbon oxides, hydrocarbons) when heated to decomposition.

Hazardous Reactions Polymerization is not expected to occur.

11. TOXICOLOGICAL INFORMATION

Health Hazard Summary	Toxic - narcotic. Use safe work practices to avoid eye-skin contact and vapour inhalation. Over exposure may result in anaemia, loss of appetite, tremors and blood, liver and kidney damage. Occupational exposure to n-hexane may cause peripheral neuropathy (nerve damage) in workers, with numbness or tingling in extremities. Recovery from effects on the peripheral nervous system is not immediate upon cessation of exposure, and effects may progress 2-3 months.
Eye	Irritant. Contact may result in irritation, lacrimation, pain, redness and conjunctivitis. Prolonged contact may result in corneal burns and possible permanent damage.
Inhalation	Irritant - toxic. Over exposure may result in irritation of the nose and throat, headache, fatigue, loss of appetite, nausea and vomiting. At high levels; dizziness, breathing difficulties, pulmonary oedema and unconsciousness. Chronic exposure may cause liver, kidney and CNS damage.
Skin	Irritant. Contact may result in drying and defatting of the skin, rash and dermatitis. May be absorbed through skin with toxic effects.
Ingestion	Toxic. Ingestion may result in nausea, vomiting, abdominal pain, dizziness, fatigue and diarrhoea. Ingestion of large quantities may result in liver and kidney damage, and unconsciousness. Aspiration may result in chemical pneumonitis and pulmonary oedema.
Toxicity Data	N-HEXANE (110-54-3) LC50 (Inhalation): > 5 mg/L LD50 (Ingestion): 28.7 g/kg (rat) ACETONE (67-64-1) LC50 (Inhalation): 44000 mg/m³/4 hours (mouse) LD50 (Ingestion): 3000 mg/kg (mouse) LD50 (Skin): > 9400 uL/kg (guinea pig) TOLUENE (108-88-3) LC50 (Inhalation): 400 ppm/24 hours (mouse) LD50 (Ingestion): 636 mg/kg (rat) LD50 (Skin): 14100 uL/kg (rabbit)

12. ECOLOGICAL INFORMATION

Environment	Limited ecotoxicity data was available for this product at the time this report was prepared. Ensure appropriate measures are taken to prevent this product from entering the environment.
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13. DISPOSAL CONSIDERATIONS

Waste Disposal	Wearing the protective equipment outlined, ensure all ignition sources are extinguished. For small quantities, absorb on paper, sand or similar and evaporate under a fume cupboard or open area. For large volumes, atomise into incinerator (mixing with more flammable solvent if required) or recycle by gravimetric separation, distilling & reusing. Contact the manufacturer for additional information if required.
Legislation	Dispose of in accordance with relevant local legislation.

14. TRANSPORT INFORMATION



CLASSIFIED AS A DANGEROUS GOOD BY THE CRITERIA OF THE ADG CODE

Shipping Name	ADHESIVES containing flammable liquid			
UN No.	1133	DG Class	3	Subsidiary Risk(s) None Allocated
Pkg Group	II	Hazchem Code	3[Y]E	EPG 3A1

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IATA

Shipping Name	ADHESIVES containing flammable liquid			
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Pkg Group	II			

IMDG

Shipping Name	ADHESIVES containing flammable liquid			
UN No.	1133	DG Class	3	Subsidiary Risk(s) None Allocated
Pkg Group	II			

15. REGULATORY INFORMATION

Poison Schedule Classified as a Schedule 5 (S5) Poison using the criteria in the Standard for the Uniform Scheduling of Drugs and Poisons (SUSDP).

AICS All chemicals listed on the Australian Inventory of Chemical Substances (AICS).

16. OTHER INFORMATION

Additional Information SYNERGISM - ANTAGONISM: Ingredients in this product may act together to aggravate or reduce adverse effects. Accordingly the time weighted average concentration (TWA) provided for single ingredients should be considered as a guide only and all due care exercised when handling.

WORK PRACTICES - SOLVENTS: Organic solvents may present both a health and flammability hazard. It is recommended that engineering controls should be adopted to reduce exposure where practicable (for example, if using indoors, ensure explosion proof extraction ventilation is available). Flammable or combustible liquids with explosive limits have the potential for ignition from static discharge. Refer to AS 1020 (The control of undesirable static electricity) and AS 1940 (The storage and handling of flammable and combustible liquids) for control procedures.

ABBREVIATIONS:

ADB - Air-Dry Basis.

BEI - Biological Exposure Indice(s)

CAS# - Chemical Abstract Service number - used to uniquely identify chemical compounds.

CNS - Central Nervous System.

EINECS - European INventory of Existing Commercial chemical Substances.

IARC - International Agency for Research on Cancer.

M - moles per litre, a unit of concentration.

mg/m3 - Milligrams per cubic metre.

NOS - Not Otherwise Specified.

NTP - National Toxicology Program.

OSHA - Occupational Safety and Health Administration.

pH - relates to hydrogen ion concentration using a scale of 0 (high acidic) to 14 (highly alkaline).

ppm - Parts Per Million.

RTECS - Registry of Toxic Effects of Chemical Substances.

TWAVES - Time Weighted Average or Exposure Standard.

HEALTH EFFECTS FROM EXPOSURE:

It should be noted that the effects from exposure to this product will depend on several factors including: frequency and duration of use; quantity used; effectiveness of control measures; protective equipment used and method of application. Given that it is impractical to prepare a Chem Alert report which would encompass all possible scenarios, it is anticipated that users will assess the risks and apply control methods where appropriate.

PERSONAL PROTECTIVE EQUIPMENT GUIDELINES:

The recommendation for protective equipment contained within this Chem Alert report is provided as a guide only. Factors such as method of application, working environment, quantity used, product concentration and the availability of engineering controls should be considered before final selection of personal protective equipment is made.

COLOUR RATING SYSTEM: RMT has assigned all Chem Alert reports a colour rating of Green, Amber or Red for the sole purpose of providing users with a quick and easy means of determining the hazardous nature of a product. Safe handling recommendations are provided in all Chem Alert reports so as to clearly identify how users

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can control the hazards and thereby reduce the risk (or likelihood) of adverse effects. As a general guideline, a Green colour rating indicates a low hazard, an Amber colour rating indicates a moderate hazard and a Red colour rating indicates a high hazard.

While all due care has been taken by RMT in the preparation of the Colour Rating System, it is intended as a guide only and RMT does not provide any warranty in relation to the accuracy of the Colour Rating System. As far as is lawfully possible, RMT accepts no liability or responsibility whatsoever for the actions or omissions of any person in reliance on the Colour Rating System.

Report Status

This document has been compiled by RMT on behalf of the manufacturer of the product and serves as the manufacturer's Material Safety Data Sheet ('MSDS').

It is based on information concerning the product which has been provided to RMT by the manufacturer or obtained from third party sources and is believed to represent the current state of knowledge as to the appropriate safety and handling precautions for the product at the time of issue. Further clarification regarding any aspect of the product should be obtained directly from the manufacturer.

While RMT has taken all due care to include accurate and up-to-date information in this MSDS, it does not provide any warranty as to accuracy or completeness. As far as lawfully possible, RMT accepts no liability for any loss, injury or damage (including consequential loss) which may be suffered or incurred by any person as a consequence of their reliance on the information contained in this MSDS.

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End of Report